

UNIVERSITY OF THE ARMED FORCES - ESPE

Department of Exact Sciences

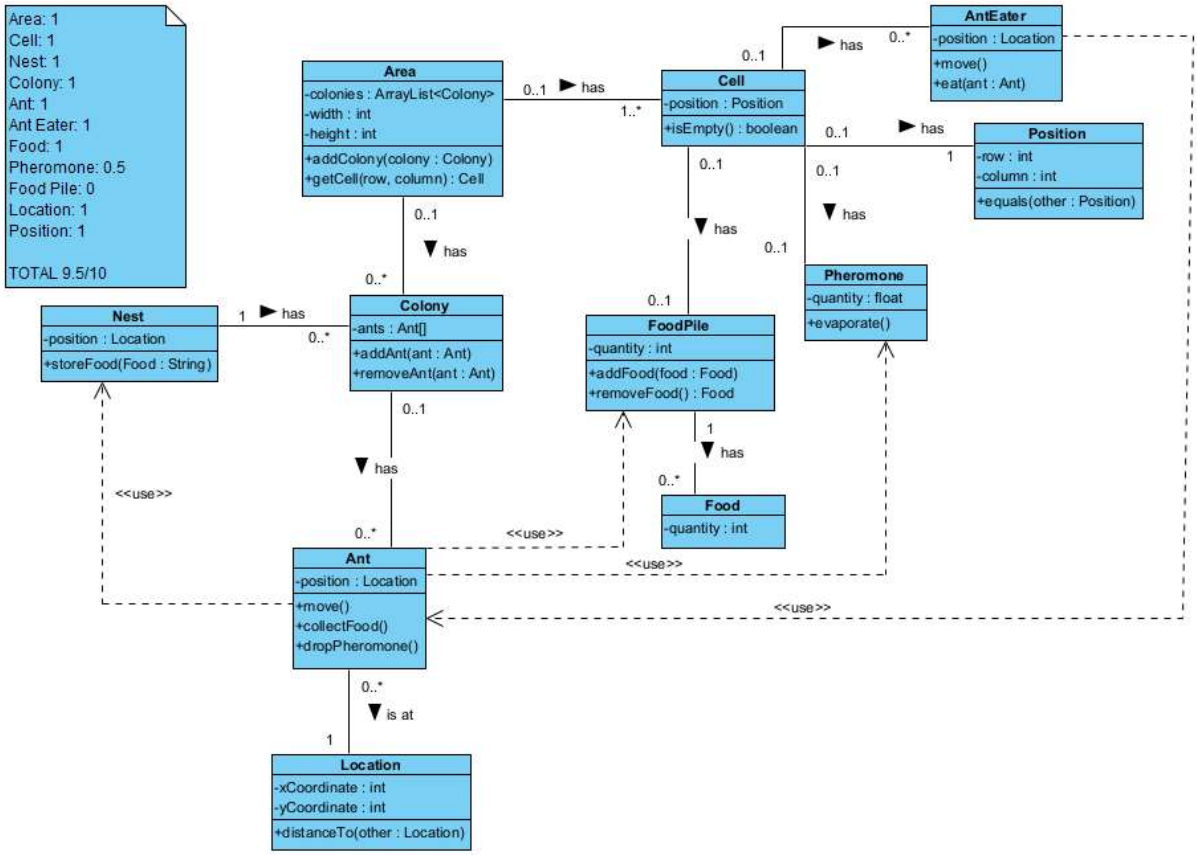
Object - Oriented Programming

Workshop11-FindingObjectsAntsAndAntEaters

Verónica Daniela Ruiz Gámez

Jorge Edison Lascano

October 16, 2026


Area: 1
Cell: 1
Nest: 1
Colony: 1
Ant: 1
Ant Eater: 1
Food: 1
Pheromone: 0.5
Food Pile: 0
Location: 1
Position: 1
TOTAL 9.5/10
Area
-colonies : ArrayList<Colony>
-width : int
-height : int
+addColony(colony : Colony)
+getCell(row, column) : Cell
Cell
-position : Position
+isEmpty() : boolean
AntEater
-position : Location
+move()
+eat(ant : Ant)
Position
-row : int
-column : int
+equals(other : Position)
Nest
-position : Location
+storeFood(Food : String)
Colony
-ants : Ant[]
+addAnt(ant : Ant)
+removeAnt(ant : Ant)
FoodPile
-quantity : int
+addFood(food : Food)
+removeFood() : Food
Pheromone
-quantity : float
+evaporate()
Food
-quantity : int
Ant
-position : Location
+move()
+collectFood()
+dropPheromone()
Location
-xCoordinate : int
-yCoordinate : int
+distanceTo(other : Location)
has
is at
<<use>>